

VALLEY VIEW UNIVERSITY
DEPARTMENT OF COMPUTING SCIENCE
Final Year Research Projects I & II
Submission Guidelines

A. INTRODUCTION

The Department has adopted two categories of research projects which we call

- i. Research and System Development
- ii. Research Only [with simulation]

REQUIRED CHAPTERS

▪ Research and System Development Required Chapters

1. General Introduction (Research Proposal)
2. Literature Review
3. Crystallization of the Research Problem
4. Analysis of the Proposed System
5. Detailed design of the Proposed System
6. System Implementation and Testing
7. System Documentation
8. Conclusion and Recommendations
- References

▪ Research Only Required Chapters

1. General Introduction (Research Proposal)
2. Literature Review
3. Research Methodology
4. Findings and Discussions
5. Framework or Model Development
6. Testing of Framework or Model with Existing Tools
7. Conclusion and Recommendations
- References

B. DOCUMENT DETAILS (Research and System Development)

Cover Page

Table of Contents

Acknowledgements

Declaration

Abstract

Besides the Cover Page you are required to number pages of these in Roman Numerals

1. General Introduction (Research Proposal)

- 1.0 Introduction
- 1.1 Subject and Field of Study
- 1.2 Study Objectives
 - 1.2.1 Global (General) Objectives
 - 1.2.2 Specific Objectives
- 1.3 Background to the Study
- 1.4 Scope of Study
- 1.5 Justification of the Study/Significance of Study
- 1.6 Methodology (*Interview, Questionnaire, Observation, Desktop Review*) **Please Note:** For Research + System Development, specify and justify the Software Development Process Model to be employed.
- 1.7 Expected Results of the Study & Possible Use (Research/Study Deliverables)
- 1.8 Presentation of Thesis (Chapter-by-Chapter Summary of Thesis)
- 1.9 Study Work Plan (Timelines) **[10 Marks]**

2. Literature Review

You are required to conduct a critical and in depth evaluation of what have been published on your chosen research area/topic. The purpose of this exercise is to summarize, synthesize and analyze the arguments of others to support your own argument. Therefore, describe and analyze the knowledge that exists and identify the gaps in research related to your project topic. This should clarify the relationship between your research topic and the work that has previously been done.

Also, you are required to review at least two existing computer-based systems for those undertaking Research and System Development. **Please note:** Your website references should not exceed 30% of the total references. **[30 Marks]**

3. Crystallization of the Problem

You are required to undertake a study of the system implemented by your case study organization(s). Provide information on the following i.

- Background/History
- ii. How they operate (components/features of the system they use even if it is manual, processes or procedures involved). Include the system context diagram and the process or data flow diagrams.
- iii. The strength/advantages and the weaknesses/disadvantages of the system they use.

Please note: The list above is not exhaustive. You can discuss with your supervisor on additions to include. **[20 Marks]**

4. Analysis of the Proposed System

A typical analysis of the proposed system has the following subheadings;

4.0 Overview of the proposed system

- Functional Requirements
 - Input requirement
 - Processing requirements
 - Output requirements
- Non-functional Requirements
 - Usability
 - Security etc.

4.1 Major features/components of the proposed system

4.2 Benefits/Advantages of the proposed system

4.3 System Context Diagram of the proposed system

Please note: The list above is not exhaustive. You can discuss with your supervisor on additions to include. **[20 Marks]**

5. Detailed design of the Proposed System

5.0 Functional Processes of the Proposed System

5.1 Algorithm and flowchart of the processes

5.2 Data Flow Diagrams

5.3 Data Dictionary

- Database Schema
- Tables
- Table Relationship Diagram
- Entity relationship diagrams
- Use Cases or User Scenarios
- UML Class Diagram

6. System Implementation & Testing

6.0 Implementation

- Description of both the Hardware and Software required for the implementation (installation) of the proposed system

6.1 Testing

- Description of the testing strategy
- Statement of Test Cases for the proposed system

6.2 Sample codes

-

7. System Documentation

- User Manual for the system

Provide both text and flowchart of how to use the various components of the system.

8. Conclusion and Recommendation

C. DOCUMENT DETAILS (Research Only-with simulation)

Cover Page Table of Contents Acknowledgements Declaration Abstract

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2. Literature Review

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Please note: Your website references should be less than 30% of the total references. **[30 Marks]**

3. Research Methodology

[20 Marks]

- 3.1 Type of Research
- 3.2 Population
- 3.3 Sample
- 3.4 Sampling technique
- 3.5 Data collection method
- 3.6 Type of data to be collected
- 3.7 Instrument for data collection
- 3.8 Instrument validity and reliability
- 3.9 Instrument structure to meet research objectives
- 3.10 Method of data analysis
- 3.11 Instrument (*i.e. Questionnaire, experiment design, interview questions, etc*)
- 3.12 Computer Simulation of Instrument **[20 Marks for the questionnaire ONLY]** **Here if you are developing a machine learning model you provide the details of the machine learning algorithm(s) you are using (assumptions, learning criteria, optimization algorithms, model predictions, dataset, etc).**

4. Data Presentation, Findings and Discussions
5. Framework or Model Development
6. Testing of Framework or Model with Existing Tools and dataset
7. Conclusion and Recommendations

D. REFERENCES/BIBLIOGRAPHY

You are required to follow the format below

- **Books**

Author(s), *Title*, Publisher(s), Edition, Year of Publication

Example: Quarshie H, *Computer Science Research Made Simple*, 1st ed, 2011

- **Journals Article**

Author(s), *Title*, Journal Name, Year of Publication, Pages

Example: Quarshie H, *Computer Science Research Made Simple*, Journal for Advances in Computer Technology, 2011, Pages (34 – 37)

- **Conference Proceedings/Report**

Author(s), *Title*, Pages, Year of Publication, Name of Conference, City, Publisher(s)

Example: Quarshie H, Damoah D., *Computer Science Research Made Simple*, Pages (34 – 37), 2011, Ghana Researchers Conference, Koforidua, Silverplatter Ventures

- **Document/information from Website**

Author(s), Name of Web Page, Name of Website, Year of Publication, *URL*

Example: Silverplatter Ventures, What is Literature Review, Experiment Resources, 2011, <http://www.experiment-resources.com/what-is-a-literaturereview.html>

- **Interview**

Interviewee, Title, Interviewer, Date

Example: Mansa Mansotwenee, Operations of Silverplatter Ventures, Quarshie H, 11-11-2011

NOTE: Sort your references in alphabetical order.

E. GENERAL DOCUMENT FORMAT

The general format of the document you present are as follows;

- **Font Style:** - Courier New
- **Font Size:** - 12
- **Paragraph Line Spacing:** - Double Line spacing
- **Margins:** - 1 inch for Top, Bottom, Left, and Right
- **Text Alignment:** - Justify
- **In text referencing:** - APA Format
- **Page Numbers:** - Should be centered at the bottom page. Avoid the use of page numbers such as

- No footers or headers.

- Labels for all figures should be below the figure and the figure should be broad enough (i.e. cover at least $\frac{1}{2}$ of the page).
- **Please Note:** We will look out for the general presentation layout, and discussion flow of the report.
- You should have ONLY one introduction, and that should be in Chapter 1.

COMPUTING SCIENCES DEPT

F. GRADING SCHEDULE

**VALLEY VIEW UNIVERSITY
INSTITUTE OF COMPUTER SCIENCE
FINAL YEAR PROJECT I GRADING SCHEDULE**

STUDENT NAME:

STUDENT ID #:

PROJECT TITLE:

DESCRIPTION	MARKS ALLOTTED	MARKS ASSIGNED
<u>Defining the purpose</u> of the research, justifying the research work and clearly stating how the research was conducted and the summary of the presentation of the research report	10 Marks	
<u>Literature Review</u> - The extent, scope, details and relevance of the Literature Review of previous research and other works conducted in the area of the project.	30 Marks	
<u>Crystallization of the Problem</u> – Demonstration of the understanding of the problem to be solved.	20 Marks	
<u>Analysis</u> - The relevance, scope, and the technical details, the suitability of the adopted methodology and demonstration of the student's comprehension of the various technical, conceptual and methodological issues pertaining to the analysis conducted.	20 Marks	
<u>General Presentation</u> laid out, discussion flow of the report.	10 Marks	
<u>Oral Presentation</u>	10 Marks	
TOTAL	100	

Name of Supervisor/Moderator: Signature:

Grade:..... Date:

**VALLEY VIEW UNIVERSITY
INSTITUTE OF COMPUTER SCIENCE
FINAL YEAR PROJECT II GRADING SCHEDULE**

STUDENT NAME:

STUDENT ID#:

PROJECT TITLE:

DESCRIPTION	MARKS ALLOTTED	MARKS ASSIGNED
<u>Design</u> – The relevance, scope, and the technical details, the suitability of the adopted design tools used and demonstration of the student’s comprehension of the various technical, issues pertaining to the design of the proposed system. The relevance and completeness of the system design outputs.	50	
<u>Implementation</u> – The extent to which the implemented system addresses the key specific objectives of the project. The demonstration of the working features of the developed system in terms of sample data/information. The quality and the relevancy of the user manual produced. The system documentation and the evidence of testing of the system.	25	
<u>General Presentation</u> laid out discussion flow of the report.	15	
<u>Oral Presentation</u>	10	
<u>Total</u>	100	

Name of Supervisor/Moderator: Signature:

Grade:

Date:

APPENDIX A – Cover Page Sample

VALLEY VIEW UNIVERSITY [line 1]
FACULTY OF SCIENCE [line 2]
DEPARTMENT OF COMPUTING SCIENCES &
ENGINEERING [line 3]

A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF [line 11] THE
REQUIREMENTS FOR THE BACHELOR OF SCIENCE (BSC.) IN _____
DEGREE

TOPIC: [line 25]
[sample] A COMPARATIVE STUDY OF THE EFFECT OF Facebook ON THE
FACIAL EXPRESSIONS OF COMPUTER SCIENTISTS

BY: [line 35]
KWAME MUSA MANSOTWENEE

ID: [line 39]
299CT01067071

SUPERVISOR: [line 43] PROF. PEACE ASOMDWEHENE

DATE: [line 50]
MAY 2012

Please Note:

- **Text Alignment:** Centered
- **Paragraph Line Spacing:** - Double Line spacing
- **Font Size:** 12
- **Font Style:** Courier New

Bold the entire text for the cover page No underline or italicize

APPENDIX B: Sample Declaration

DECLARATION

This is to declare that, the research work underlying this thesis has been carried out by the under mentioned student under the supervisor. Both student and the supervisor certify that the work documented in this thesis is the output of the research conducted by the student as part of the final year project work in partial fulfillment of the requirement of the Bachelor of Science in Information Technology degree.

STUDENT:

Kwaku Troublesome

SIGNATURE

DATE:

SUPERVISOR:

Prof. Chaos Peace

SIGNATURE

DATE: